Notes 1: "What is a (mathematical) model?"

This video explains what a mathematical model is and why we use one.

1. What is a Model?

In statistics, a **model** is not a physical object but a simplified, mathematical way to describe a **relationship** between variables.

- **Goal:** To understand how one or more *input variables* (also called *predictors* or *independent variables*) can help explain or predict an *output variable* (also called a *response* or *dependent variable*).
- **Video Example:** Modeling mouse size.
  - **Output Variable:** `Mouse Size`
  - **Input Variable:** `Mouse Weight`
- The model describes the relationship: "As `Mouse Weight` increases, `Mouse Size` also tends to increase".

2. How is a Model Represented?

A model is most often represented as a **mathematical equation**.

- If you plot your data (e.g., weight on the x-axis, size on the y-axis), the model is a **line or curve** that tries to get as close as possible to all the data points.
- A simple linear model (a straight line) would have an equation like:
  `Mouse Size = (some number * Mouse Weight) + (another number)`
  (This is the familiar `y = mx + b` formula for a line).

3. What Are Models Used For?

1. **Exploring Relationships:** Models help us formally describe and understand *how* variables are connected.
2. **Making Predictions:** This is a primary use. Once you have a model, you can use it to predict the output for new, unseen data.
   - *Example:* If you find a new mouse that weighs 4 units, you can plug "4" into your equation to *predict* its size (the video predicts 3.3 units).

4. Key Concept: Models are Approximations

A model is a *simplification* of reality. It is **not perfect**.

- The real data points will be scattered *around* the model's line; they will not all fall perfectly *on* the line.
- The distance from a real data point to the model's line is called the **error** or **residual**.
- A large part of statistics is dedicated to finding the model that has the *least amount of total error* and determining if that model is a **good approximation** of the data.

5. Types of Models

Models can be simple or very complex:

- **Simple Linear Model:** A straight line relating one input to one output (e.g., `Weight -> Size`).
- **Non-Linear Model:** A *curve* used when the relationship isn't a straight line.
  - *Video Example:* The relationship between `Drug Dose` and `Hair Growth`. At first, more drug causes more growth, but eventually, the effect *plateaus* (flattens out), and more drug doesn't help.
- **Multivariable Model:** A more complex model that uses *multiple inputs* to predict an output.
  - *Video Example:* `(Gene X Expression) + (Gene Z Expression) -> Mouse Size`

Summary: A model is a mathematical equation that describes the relationship between variables. We use it to make predictions, and we use statistics to determine how useful and reliable that model is.

Notes 2: "Hypothesis Testing and The Null Hypothesis"

This video explains the core logic behind hypothesis testing, focusing on the most important concept: the **Null Hypothesis**.

1. The Core Problem: Is a Difference Real or Just Random?

When we do an experiment, we always see *some* difference. The central question is: "Is the difference we see *real* (due to our experiment) or is it just due to *random chance*?"

- **Example:** Testing `Drug A` vs. `Drug B` for virus recovery.
- **The Role of Random Variation:** Even if you give the *same drug* to three people, they will *not* recover in the exact same amount of time.
- This is because of **uncontrollable factors** (natural variation) like genetics, diet, job stress, exercise, etc.. This variation is "noise" in the data.
- **The Question:** We see that, on average, `Drug A` was 15 hours faster than `Drug B`. Is this 15-hour difference *real*, or is it just due to the random "noise" (e.g., the healthy people just happened to get Drug A)?

2. The Flaw of Testing a Specific Hypothesis

It's tempting to see the 15-hour difference and form this hypothesis: "The true difference between Drug A and B is 15 hours".

- **The Problem:** This specific number (15) was just the result of *one* experiment.
- If you repeat the experiment, you will get *different* results (e.g., 12 hours, 13.5 hours, etc.) due to random variation.
- It's impractical and illogical to test an infinite number of possible hypotheses (Is the difference 13? Is it 12? Is it 13.5?).

3. The Solution: The Null Hypothesis (H_0)

Instead of testing a specific, random value (like "15 hours"), we test a single, fixed, meaningful value: **zero**.

- The **Null Hypothesis** (symbolized as H_0) is the default assumption, or the "skeptical" position.
- **H_0 is the hypothesis of "no effect" or "no difference"**.
- **For our example, H_0 is:** "There is **no difference** in average recovery time between Drug A and Drug B. The true difference is 0 hours.".

4. The "Innocent Until Proven Guilty" Framework

Think of hypothesis testing as a court trial:

- **H_0 (Null Hypothesis):** The defendant is **innocent**. (This is the default position).
- **H_A (Alternative Hypothesis):** The defendant is **guilty**. (This is what the researcher often hopes to prove. The video teases this for the next lesson).
- **The Data:** This is your **evidence**.

The goal is **not** to *prove* the defendant is guilty (H_A). The goal is to see if you have *enough evidence* (data) to **reject** the "innocent" (H_0) claim.

5. The Two Possible Outcomes of a Test

1. **Reject the Null Hypothesis (H_0)**
   - This is like the jury saying "Guilty!"
   - It means the difference you observed in your data is so large that it is **highly unlikely** to have occurred by random chance alone.
   - *Example:* If you test 1,000 people and Drug A is *still* 15 hours faster, that's too much of a coincidence to be random.
   - **Conclusion:** You have **statistically significant** evidence that there *is* a real difference between the drugs.
2. **Fail to Reject the Null Hypothesis (H_0)**
   - This is like the jury saying "Not Guilty."
   - **This does NOT mean the defendant is innocent (H_0 is true)!** It just means there *wasn't enough evidence to convict*.
   - It means the difference you saw was small enough that it *could* have been caused by random variation.
   - *Example:* A tiny 0.5-hour difference that flips to a -0.25-hour difference in another small test. This is just random noise.
   - **Conclusion:** You **do not** have enough evidence to say there is a difference.

Summary: We start by assuming the "boring" **Null Hypothesis** (H_0: no difference). We then collect data (evidence) and use statistics to decide if our evidence is strong enough to *reject* that "no difference" idea.

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